

Robust Joint Angle-Range-Velocity Estimation for AFDM-Based MIMO-ISAC Systems

Jie Li, Peng Xu, Kaiwen Shi, Ziyu Dong

College of Electrical and Information Engineering, Nanjing University of Aeronautics and Astronautics, Nanjing, China

Email: xupeng_eie@nuaa.edu.cn, lijie_evelyn@outlook.com

Abstract—Affine frequency division multiplexing (AFDM) is a robust alternative for high-mobility integrated sensing and communications (ISAC) systems, yet research on multiple-input multiple-output (MIMO)-AFDM echo processing remains limited due to complex multidimensional coupling. In this paper, we propose a novel joint angle–range–velocity estimation framework for monostatic MIMO-AFDM ISAC systems. We first establish a rigorous 3D discrete affine Fourier (DAF)-domain signal model. To resolve the AFDM-induced multidimensional coupling, we devise a beamspace delay-compensation and matched normalization scheme. An FFT-based GLRT/ML readout and noncoherent accumulation are then applied to form a 3D power cube. Robust target detection is achieved using 3D CA-CFAR with non-maximum suppression, followed by quadratic interpolation for precise sub-grid refinement. Simulations validate the proposed method, demonstrating superior angle–range–velocity root-mean squared error (RMSE) performance compared to existing state-of-the-art techniques and uncompensated baselines.

Index Terms—ISAC, AFDM, MIMO, delay-compensation, joint parameter estimation.

I. INTRODUCTION

Waveform design plays a central role in integrated sensing and communications (ISAC), as it dictates sensing resolution, communication reliability, and implementation complexity [1], [2]. Although orthogonal frequency division multiplexing (OFDM)-based ISAC has achieved notable progress [3]–[5], future scenarios feature high mobility and high carrier frequencies, where doubly selective channels (DSCs) degrade OFDM performance [6]–[9].

Emerging candidates like orthogonal time frequency space (OTFS) and orthogonal chirp-division multiplexing (OCDM) improve DSCs robustness but typically suffer from high complexity or restrictive receiver assumptions [10]–[12]. Recently, affine frequency division multiplexing (AFDM) has attracted interest due to its ability to achieve full diversity in DSCs by turning DAFT-based chirp parameters. With its one-dimensional modulation structure and low pilot overhead, AFDM is an especially appealing waveform for ISAC in high-mobility environments [13]–[15].

Moreover, multiple-input multiple-output (MIMO) introduces crucial spatial degrees of freedom to ISAC. Although

channel estimation and detection for multi-antenna AFDM has been investigated [16]–[18], the corresponding *echo processing* for sensing is still in its infancy. Existing high-resolution algorithms (e.g., tensor/subspace methods) frequently require stringent identifiability conditions, which undermines their engineering robustness [19]. This necessitates the development of a practical parameter estimation framework attuned to the unique properties of AFDM echoes.

In this paper, we investigate echo signal processing for MIMO-AFDM ISAC systems and develop a joint angle-range-velocity estimation framework that explicitly addresses the AFDM-induced coupling in the echo domain. The main contributions are summarized as follows:

- We formulate a rigorous DAF-domain signal model for monostatic MIMO-AFDM ISAC. Leveraging the DAFT processing chain, we organize the received echoes into a 3D sensing cube indexed by receive element, subcarrier, and symbol.
- To resolve the inherent multidimensional coupling in AFDM, we propose a beamspace delay-compensation method. This approach leverages beamspace projection for signal-to-noise ratio (SNR) gain and fast-time compensation to extract the slow-time Doppler phase, facilitating an FFT-based generalized likelihood ratio test/maximum likelihood (GLRT/ML) Doppler readout via matched normalization.
- We develop a joint angle–range–velocity estimation algorithm. By applying slow-time FFT and non-coherent integration, a 3D power cube is constructed. Targets are detected using 3D CA-CFAR and non-maximum suppression, with estimation accuracy further enhanced quadratic sub-grid interpolation.
- Extensive simulations validate the efficacy of the proposed framework, demonstrating superior root-mean squared error (RMSE) performance for angle-range-velocity estimation over existing benchmarks and uncompensated schemes.

II. SYSTEM MODEL

We consider a monostatic MIMO-AFDM ISAC station (BS) that employs co-located uniform linear arrays (ULAs) with M_t transmit and G receive elements. The BS maps pilots and

This work was supported in part by the National Science and Technology Major Project 2026ZD1306000, and in part by the National Natural Science Foundation of China under Grants 62427801 and 62231015. (Corresponding author: Jie Li.)

quadrature phase shift keying (QPSK) data onto the DAF-domain lattice and transmits AFDM waveforms to K single-antenna users while probing P point targets.

A. AFDM Transmit Signal and Coupling Analysis

Let N_c denote the number of chirp-subcarriers with spacing Δf , bandwidth $B = N_c \Delta f$, and sampling period $T_s = 1/B$. Each AFDM symbol has duration $T_{\text{sym}} = (N_c + N_{\text{cp}})T_s$, where N_{cp} is the CPP length. Denote the DAF-domain transmit block by $\mathbf{X} \in \mathbb{C}^{N_c \times N_s \times M_t}$ and define $\mathbf{x}_m[k] = [X(0, k, m), \dots, X(N_c - 1, k, m)]^T \in \mathbb{C}^{N_c}$ as the DAF-domain vector for antenna $m \in \{1, \dots, M_t\}$ at symbol $k \in \{0, \dots, N_s - 1\}$.

The n -th time-domain sample ($n = 0, \dots, N_c - 1$) of the k -th symbol at the m -th transmit antenna is generated by the N_c -point inverse DAFT (IDAFT):

$$s_{m,k}[n] = \frac{1}{\sqrt{N_c}} \sum_{p=0}^{N_c-1} X(p, k, m) e^{j2\pi(-c_1 n^2 + \frac{np}{N_c} - c_2 p^2)}, \quad (1)$$

where $c_1 = \frac{2(\alpha_{\text{max}} + k_v) + 1}{2N_c}$ and $c_2 \ll \frac{1}{2N_c}$ are the AFDM chirp parameters. Or in matrix form for the m -th antenna:

$$\mathbf{s}_{m,k} = \mathbf{\Lambda}_{c_1}^H \mathbf{F}^H \mathbf{\Lambda}_{c_2}^H \mathbf{x}_m[k], \quad (2)$$

where $\mathbf{F} \in \mathbb{C}^{N_c \times N_c}$ is the unitary N_c -point DFT matrix with $[\mathbf{F}]_{n,p} = \frac{1}{\sqrt{N_c}} e^{-j\frac{2\pi}{N_c} np}$, and $\mathbf{\Lambda}_c$ is the diagonal chirp matrix with $[\mathbf{\Lambda}_c]_{n,n} = e^{j2\pi cn^2}$.

Let $x[n]$ be a length- N_c sequence and define its DAFT as

$$X_{\text{DAF}}[p] = \frac{1}{\sqrt{N_c}} \sum_{n=0}^{N_c-1} x[n] e^{-j2\pi(c_1 n^2 + \frac{np}{N_c} + c_2 p^2)}. \quad (3)$$

Consider a pure delay by ℓ samples, $x_\ell[n] = x[(n-\ell)_{N_c}] e^{j2\pi c_1(N_c^2 - 2N_c n)}$ for $n < 0$ as enforced by the chirp-periodic prefix (CPP). A direct index substitution gives

$$X_{\text{DAF},\ell}[p] = e^{-j\frac{2\pi}{N_c}\ell p} e^{-j2\pi c_2(p^2 - (p-\ell)^2)} X_{\text{DAF}}[(p-\ell)_{N_c}]. \quad (4)$$

Likewise, a Doppler f_d multiplies the time waveform by $e^{j2\pi f_d t}$ and, at the symbol level, produces a *linear phase* across the slow-time index k :

$$x_k[n] \mapsto x_k[n] e^{j2\pi f_d k T_{\text{sym}}} \Rightarrow e^{jk \omega_v(v)}, \quad (5)$$

$$\omega_v(v) = \frac{4\pi f_c T_{\text{sym}}}{c} v.$$

A delay by ℓ samples produces a cyclic shift $p \mapsto (p - \ell) \bmod N_c$ along the DAF-domain axis with a deterministic phase, cf. (4); a Doppler f_d induces a linear phase ramp across the slow-time index k (with $\omega_v(v)$ defined in (5)).

B. Echo Model and 3D Data Cube

We adopt a far-field monostatic setting with co-located Tx/Rx ULAs of spacings $d_t = d_r = \lambda/2$. The target is described by its angle θ_q , range R_q , radial velocity v_q , and complex reflectivity γ_q . The associated round-trip delay and Doppler are $\tau_q = 2R_q/c$ and $f_{d,q} = 2v_q/\lambda$, respectively,

where f_c is the carrier frequency, c is the speed of light, and $\lambda = c/f_c$ is the wavelength. The receive steering vector is

$$\mathbf{a}_R(\theta) = [1, e^{j\omega_a(\theta)}, \dots, e^{j(G-1)\omega_a(\theta)}]^T, \quad (6)$$

where $\omega_a(\theta) = -2\pi(d_r/\lambda) \sin \theta$. Similarly, the Tx ULA has steering vector $\mathbf{a}_T(\theta) \in \mathbb{C}^{M_t}$. After CPP removal and DAFT/FFT operations, the processed echo at receive element g , subcarrier p , and symbol k is

$$y(g, p, k) = \sum_{q=1}^P \gamma_q a_{R,g}(\theta_q) \mathbf{a}_T^H(\theta_q) \mathbf{x}_p[k] e^{j(p\omega_r(R_q) + k\omega_v(v_q))} + z(g, p, k), \quad (7)$$

where $z(g, p, k)$ is the additive white gaussian noise and

$$\omega_r(R) = -\frac{4\pi\Delta f}{c} R, \quad \omega_v(v) = \frac{4\pi T_{\text{sym}} f_c}{c} v. \quad (8)$$

Stacking samples yields the 3D sensing cube

$$\mathcal{Y} \in \mathbb{C}^{G \times N_c \times N_s}, \quad \mathcal{Y}(g, p, k) = y(g, p, k), \quad (9)$$

with fast-/slow-time steering vectors $\mathbf{b}(R) = [1, e^{j\omega_r(R)}, \dots, e^{j(N_c-1)\omega_r(R)}]^T$ and $\mathbf{c}(v) = [1, e^{j\omega_v(v)}, \dots, e^{j(N_s-1)\omega_v(v)}]^T$. Defining $s_q(p, k) \triangleq \mathbf{a}_T^H(\theta_q) \mathbf{x}_p[k]$, a convenient vectorized representation is

$$\text{vec}(\mathcal{Y}) = \sum_{q=1}^P \gamma_q \left(\mathbf{c}(v_q) \otimes \mathbf{b}(R_q) \otimes \mathbf{a}_R(\theta_q) \right) \circ \text{vec}(\mathcal{S}_q) + \text{vec}(\mathcal{Z}), \quad (10)$$

where $\mathcal{S}_q \in \mathbb{C}^{1 \times N_c \times N_s}$ collects $s_q(p, k)$ across (p, k) .

III. JOINT ANGLE-RANGE-VELOCITY ESTIMATION

In this section, we extract the angle-range-velocity information of potential targets from the received echo signals by fully exploiting the 3D data cube. While similar problems have been addressed in MIMO-OFDM systems, the unique delay-Doppler coupling in AFDM necessitates a modified approach.

A. Beamspace DAF-domain Data

To analyze the spatial spectrum, we introduce an angle grid $\{\theta(n_a)\}_{n_a=0}^{N_a-1}$, so $\omega_a(\theta(n_a)) = 2\pi \frac{n_a - n_0}{N_a}$, $n_0 = \lfloor \frac{N_a}{2} \rfloor$. For each pair (p, k) , the projection of the G -dimensional snapshot $\mathbf{y}(:, p, k)$ onto the angular grid is calculated as

$$Y_{n_a}(p, k) = \mathbf{a}_R^H(\theta(n_a)) \mathbf{y}(:, p, k). \quad (11)$$

Substituting the received signal model yields the angle-domain slice for target subset $\mathcal{Q}_{n_a} \subseteq \{1, \dots, P\}$ within the n_a -th beam:

$$Y_{n_a}(p, k) = \sum_{q \in \mathcal{Q}_{n_a}} \gamma_q \mathbf{a}_R^H(\theta(n_a)) \mathbf{a}_R(\theta_q) \mathbf{a}_T^H(\theta_q) \mathbf{x}_p[k] \times e^{j(p\omega_r(R_q) + k\omega_v(v_q))} + \tilde{z}(p, k; n_a). \quad (12)$$

B. Delay Compensation and Per- p Matched Filtering

Let $\ell \in \{0, 1, \dots, N_{\text{cp}} - 1\}$ denote a candidate discrete delay. The AFDM property that a delay produces a cyclic shift in p together with a deterministic phase can be countered by the diagonal operator

$$L_\ell \triangleq \text{diag}\left(e^{j\frac{2\pi}{N_c}\ell p}\right)_{p=0}^{N_c-1}. \quad (13)$$

Applying L_ℓ column-wise to $\mathbf{Y}_{n_a}$ gives the delay-compensated slice

$$\begin{aligned} Z_{n_a}^{(\ell)}(p, k) &= e^{j\frac{2\pi}{N_c}\ell p} Y_{n_a}(p, k) \\ &\approx \sum_{q \in \mathcal{Q}_{n_a}} \gamma_q s_q(p, k) e^{j(k\omega_v(v_q) + p(\omega_r(R_q) + \frac{2\pi}{N_c}\ell))} + \tilde{z}. \end{aligned} \quad (14)$$

When the trial ℓ matches the delay bin of a component with range R_q , the term in p nearly vanishes because $\omega_r(R_q) \approx -\frac{2\pi}{N_c}\ell$ (small- c_2 regime), leaving $Z_{n_a}^{(\ell)}(p, k) \approx \sum_{q \in \mathcal{Q}_{n_a}} \gamma_q s_q(p, k) e^{jk\omega_v(v_q)} + \tilde{z}$, i.e., *Doppler-only* phase along k and no systematic phase along p .

C. Slow-Time FFT and 3D Detection

To suppress residual dependence on p , we form a noncoherently integrated power cube,

$$\mathcal{P}(n_a, \ell, \nu) = \sum_{p=0}^{N_c-1} |Z_{n_a}(p, \nu; \ell)|^2, \quad (15)$$

where the noncoherent accumulation can improve robustness to residual mismatches (e.g., $\mathbf{a}_T^H(\theta(n_a))\mathbf{x}_p[k]$ and small uncompensated chirp terms).

To obtain robust candidate detections, apply a 3D CA-CFAR to $\mathcal{P}$, the local noise estimate is

$$\mu(n_a, \ell, \nu) = \frac{1}{N_{\text{tr}}} \sum_{(i,j,t) \in \mathcal{T}(n_a, \ell, \nu)} \mathcal{P}(i, j, t), \quad (16)$$

where $T(n_a, \ell, \nu) = \eta \mu(n_a, \ell, \nu)$ is the window excluding the guard region and $N_{\text{tr}} = |\mathcal{T}(n_a, \ell, \nu)|$. The CFAR threshold is

$$\mathcal{P}(n_a, \ell, \nu) \geq T(n_a, \ell, \nu) \Rightarrow \text{candidate}. \quad (17)$$

We then apply non-maximum suppression to keep strict local maxima. Sub-bin delay and Doppler are refined by quadratic interpolation around each peak (n_a, ℓ_0, ν_0) using immediate neighbors. Angle can be read on-grid via n_a or refined by a 1D spectral interpolation. The output is a peak set $\mathcal{S} = \{(n_a, \hat{\ell}, \hat{\nu})\}$ for subsequent parameter mapping.

D. Index-to-Parameter Mapping and Readout

The range and velocity are mapped from the refined delay and Doppler indices, $\hat{\ell} \in [0, N_{\text{cp}})$ and $\hat{\nu} \in [0, N_v)$, based on the grid spacing

$$\hat{R} = \frac{c\hat{\ell}}{2N_c\Delta f}, \quad \hat{v} = \frac{c\hat{\nu}}{2N_vT_{\text{sym}}f_c}. \quad (18)$$

The angle is directly read from the spatial grid index n_a :

$$\hat{\theta} = \arcsin\left(-\frac{\lambda}{d_r} \cdot \frac{n_a - n_0}{N_a}\right). \quad (19)$$

The grid sizes determine the system's unambiguous spans $(R_{\text{max}}, v_{\text{UA}})$ and parameter resolutions $(\Delta R, \Delta v, \Delta\theta)$:

$$\begin{aligned} R_{\text{max}} &= \frac{cN_{\text{cp}}}{2N_c\Delta f}, & \Delta R &= \frac{c}{2N_c\Delta f}, \\ v_{\text{UA}} &= \pm \frac{c}{4f_cT_{\text{sym}}}, & \Delta v &= \frac{c}{2N_vT_{\text{sym}}f_c}, \end{aligned} \quad (20)$$

and angular resolution $\Delta\theta \approx \lambda/(N_a d_r \cos\theta)$.

An unbiased estimate of the complex reflectivity $\hat{\gamma}$ is obtained by coherent Least Squares (LS) combining. By de-rotating the delay-compensated slice $Z_{n_a}^{(\ell)}(p, k)$ with the known transmit reference $s_{n_a}(p, k)$, we obtain:

$$\hat{\gamma} = \frac{1}{N_c N_s} \sum_{p=0}^{N_c-1} \sum_{k=0}^{N_s-1} \frac{Z_{n_a}^{(\ell)}(p, k)}{s_{n_a}(p, k) + \epsilon} e^{-j\frac{2\pi}{N_v}\hat{\nu}k}, \quad (21)$$

where $\epsilon > 0$ prevents division overflow.

The complete joint angle-range-velocity estimation procedure is summarized in **Algorithm 1**.

Algorithm 1 The Proposed Joint Angle-Range-Velocity Estimation Algorithm

- 1: **Input:** Echo cube $\mathcal{Y}$, Tx vectors $\{\mathbf{x}_p[k]\}$, angle grid, delay search ℓ , parameters N_v , CFAR params, η , ϵ .
 - 2: **Output:** Target set $\{(\hat{\theta}, \hat{R}, \hat{v}, \hat{\gamma})\}$
 - 3: **for** $n_a = 0$ **to** $N_a - 1$ **do**
 - 4: $\mathbf{Y}_{n_a}(p, k) \leftarrow \mathbf{a}_R^H(\theta(n_a)) \mathbf{y}(:, p, k)$
 - 5: $s_{n_a}(p, k) \leftarrow \mathbf{a}_T^H(\theta(n_a)) \mathbf{x}_p[k]$
 - 6: **for** $\ell = 0$ **to** $N_{\text{cp}} - 1$ **do**
 - 7: $\mathbf{Z}_{n_a}^{(\ell)} \leftarrow L_\ell \mathbf{Y}_{n_a}$
 - 8: $\tilde{Y}_{n_a}(p, k) \leftarrow Z_{n_a}^{(\ell)}(p, k) / (s_{n_a}(p, k) + \epsilon)$
 - 9: $y_{n_a}(p, k; \ell) \leftarrow \alpha_{n_a} \tilde{Y}_{n_a}(p, k)$
 - 10: **for** $p = 0$ **to** $N_c - 1$ **do**
 - 11: $Z_{n_a}(p, \nu; \ell) \leftarrow \frac{1}{\sqrt{N_v}} \sum_{k=0}^{N_s-1} y_{n_a}(p, k; \ell) e^{-j\frac{2\pi}{N_v}\nu k}$
 - 12: **end for**
 - 13: $\mathcal{P}(n_a, \ell, \nu) \leftarrow \sum_{p=0}^{N_c-1} |Z_{n_a}(p, \nu; \ell)|^2$
 - 14: **end for**
 - 15: **end for**
 - 16: **3D CA-CFAR:** Find peaks $\mathcal{S} = \{(n_a, \ell_0, \nu_0)\}$
 - 17: **Sub-grid refinement:** Calculate $(\hat{\ell}, \hat{\nu})$
 - 18: **Parameter readout:** Convert indices to $(\hat{\theta}, \hat{R}, \hat{v})$
 - 19: **Reflectivity:** Estimate $\hat{\gamma}$ using LS on compensated bins
 - 20: **Return** $\{(\hat{\theta}, \hat{R}, \hat{v}, \hat{\gamma})\}$
-

IV. NUMERICAL RESULTS

For the general setup, we assume the BS is matched with $N_c = 512$, $N_s = 64$, $N_a = 1024$, $f_c = 28 \times 10^9$ Hz and $\Delta f = 22.729 \times 10^3$ Hz. The targets are assumed to be randomly located at angles between $[-90^\circ, 90^\circ]$, ranges between $[50, 600]$ m, and radial velocities between $[-80, 80]$ m/s.

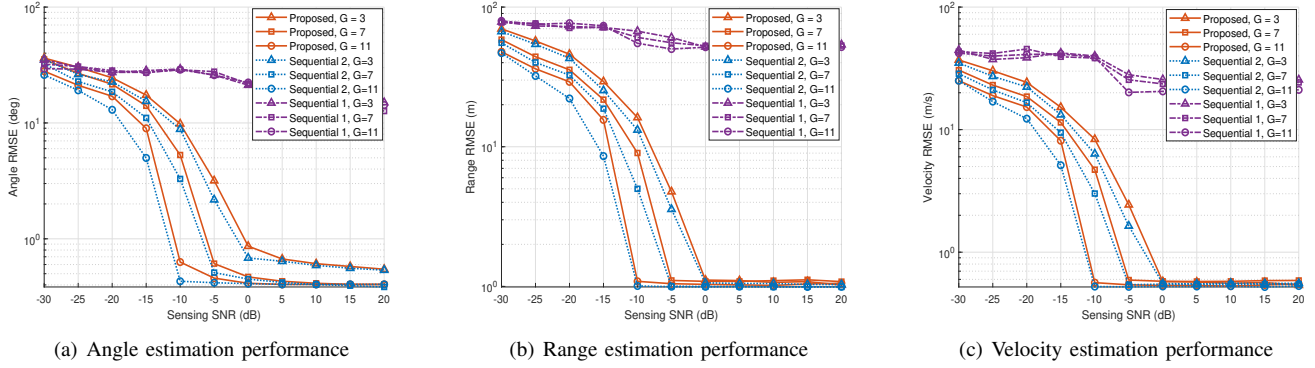


Fig. 1. Estimation performance versus SNR for different numbers of receive antennas.

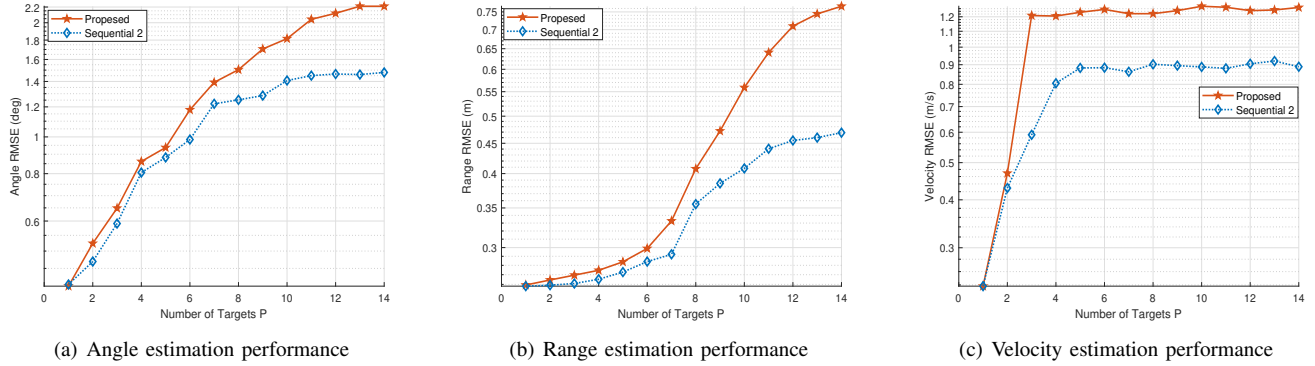


Fig. 2. Estimation performance versus the number of targets.

In addition to the proposed joint estimation algorithm, we include two baseline methods for comparison. One baseline is an estimation scheme without delay compensation, which highlights the importance of removing multidimensional data coupling, and is denoted as “Sequential 1”. The performance of a radar system employing a standard LFM CW waveform with the 3D-DFT estimation algorithm is also simulated, and is denoted as “Sequential 2”. The LFM CW waveform is known to exhibit excellent sensing performance given the same time-frequency resources [20].

We first show the RMSE versus the SNR for different numbers of receive antennas in Fig. 1. In AFDM, after the target echoes pass through the transform matrix, the delay/Doppler components experience additional phase rotations and index shifts. As a result, once the multidimensional data are superimposed, the signal becomes strongly coupled across the angle, subcarrier, and slow-time dimensions. Consequently, the joint estimation approach under the Sequential 1 setting is ineffective. Most importantly, the sensing performance achieved with the AFDM waveform is very close to that of the LFM CW radar waveform. This observation verifies the advantage of the proposed algorithm and further suggests that AFDM is a promising waveform for future ISAC systems.

In Fig. 2, we present the RMSE versus the number of targets, given an SNR of 20 dB and $G = 11$. Due to the lack of coupling cancellation in the Sequential 1 method, its

estimation error is already within the magnitude of 10^1 – 10^2 at 20 dB SNR for a single target (refer to Fig. 1), rendering it ineffective for multi-target estimation.

Conversely, while the proposed method incurs a slight performance degradation as the target count rises, it maintains performance comparable to the LFM CW single-radar benchmark. This demonstrates its capability to support satisfactory communication and sensing for ISAC applications.

V. CONCLUSIONS

This paper proposed a novel joint parameter estimation framework tailored for monostatic MIMO-AFDM ISAC systems. By deriving a rigorous 3D DAF-domain signal model, we resolve the AFDM-induced multidimensional coupling that typically obstructs echo processing in high-mobility scenarios. Specifically, we introduced a beamspace delay-compensation and matched normalization scheme that effectively decouples the range and Doppler dimensions, facilitating efficient FFT-based GLRT/ML estimation. The proposed detection pipeline-incorporating a 3D CA-CFAR with non-maximum suppression and quadratic interpolation, proved robust and precise sub-grid parameter refinement. Simulation results confirm that the proposed method significantly outperforms existing state-of-the-art techniques and uncompensated baselines in angle, range, and velocity estimation accuracy.

REFERENCES

- [1] V. Koivunen, M. F. Keskin, H. Wymeersch, M. Valkama, and N. González-Prelcic, "Multicarrier isac: Advances in waveform design, signal processing, and learning under nonidealities," *IEEE Signal Processing Magazine*, vol. 41, no. 5, pp. 17–30, 2024.
- [2] S. Wang, W. Dai, H. Wang, and G. Y. Li, "Robust waveform design for integrated sensing and communication," *IEEE Transactions on Signal Processing*, vol. 72, pp. 3122–3138, 2024.
- [3] Y. He, G. Yu, Z. Tang, J. Wang, and H. Luo, "A dual-functional sensing-communication waveform design based on ofdm," *IEEE Transactions on Wireless Communications*, vol. 23, no. 11, pp. 16962–16975, 2024.
- [4] Z. Xiao, R. Liu, M. Li, Q. Liu, and A. L. Swindlehurst, "A novel joint angle-range-velocity estimation method for mimo-ofdm isac systems," *IEEE Transactions on Signal Processing*, vol. 72, pp. 3805–3818, 2024.
- [5] S. Mura, D. Tagliaferri, M. Mizmizi, U. Spagnolini, and A. Petropulu, "Optimized waveform design for ofdm-based isac systems under limited resource occupancy," *IEEE Transactions on Wireless Communications*, vol. 24, no. 6, pp. 5241–5254, 2025.
- [6] R. Singh, A. Kaushik, W. Shin, M. D. Renzo, V. Sciancalepore, D. Lee, H. Sasaki, A. Shojaeifard, and O. A. Dobre, "Toward 6g evolution: Three enhancements, three innovations, and three major challenges," *IEEE Network*, vol. 39, no. 6, pp. 139–147, 2025.
- [7] R. Liu, L. Zhang, R. Y.-N. Li, and M. D. Renzo, "The itu vision and framework for 6g: Scenarios, capabilities, and enablers," *IEEE Vehicular Technology Magazine*, vol. 20, no. 2, pp. 114–122, 2025.
- [8] H. S. Rou, G. T. F. de Abreu, J. Choi, D. González G., M. Kountouris, Y. L. Guan, and O. Gonsa, "From orthogonal time–frequency space to affine frequency-division multiplexing: A comparative study of next-generation waveforms for integrated sensing and communications in doubly dispersive channels," *IEEE Signal Processing Magazine*, vol. 41, no. 5, pp. 71–86, 2024.
- [9] T. Wang, J. Proakis, E. Masry, and J. Zeidler, "Performance degradation of ofdm systems due to doppler spreading," *IEEE Transactions on Wireless Communications*, vol. 5, no. 6, pp. 1422–1432, 2006.
- [10] R. Hadani, S. Rakib, M. Tsatsanis, A. Monk, A. J. Goldsmith, A. F. Molisch, and R. Calderbank, "Orthogonal time frequency space modulation," in *2017 IEEE Wireless Communications and Networking Conference (WCNC)*, pp. 1–6, 2017.
- [11] M. S. Omar and X. Ma, "Performance analysis of ocdm for wireless communications," *IEEE Transactions on Wireless Communications*, vol. 20, no. 7, pp. 4032–4043, 2021.
- [12] L. Xiao, S. Li, Y. Qian, D. Chen, and T. Jiang, "An overview of otfs for internet of things: Concepts, benefits, and challenges," *IEEE Internet of Things Journal*, vol. 9, no. 10, pp. 7596–7618, 2022.
- [13] Y. Ni, Z. Wang, P. Yuan, and Q. Huang, "An afdm-based integrated sensing and communications," in *2022 International Symposium on Wireless Communication Systems (ISWCS)*, pp. 1–6, 2022.
- [14] Y. Ni, P. Yuan, Q. Huang, F. Liu, and Z. Wang, "An integrated sensing and communications system based on affine frequency division multiplexing," *IEEE Transactions on Wireless Communications*, vol. 24, no. 5, pp. 3763–3779, 2025.
- [15] S. Zhu, L. Mei, Z. Li, Z. Du, and Y. Song, "Low complexity afdm-based isac utilizing fractional delay-doppler feature extraction," *IEEE Wireless Communications Letters*, vol. 15, pp. 335–339, 2026.
- [16] H. Yin, X. Wei, Y. Tang, and K. Yang, "Diagonally reconstructed channel estimation for mimo-afdm with inter-doppler interference in doubly selective channels," *IEEE Transactions on Wireless Communications*, vol. 23, no. 10, pp. 14066–14079, 2024.
- [17] Q. Luo, J. Zhu, Z. Liu, Y. Tang, P. Xiao, G. Chen, and J. Shi, "Joint sparse graph for enhanced mimo-afdm receiver design," *IEEE Transactions on Wireless Communications*, pp. 1–1, 2025.
- [18] P. Gao, J. Zhu, K. Zhang, L. Mei, S. Ma, Y. Wang, and X. Li, "Multi-domain generalized index modulation aided mimo-afdm for future high-mobility communications," *IEEE Transactions on Vehicular Technology*, pp. 1–12, 2025.
- [19] Y. Luo, Y. L. Guan, Y. Ge, D. González G, and C. Yuen, "A novel angle-delay-doppler estimation scheme for afdm-isac system in mixed near-field and far-field scenarios," *IEEE Internet of Things Journal*, vol. 12, no. 13, pp. 22669–22682, 2025.
- [20] N. Levanon and E. Mozeson, *Radar signals*. John Wiley & Sons, 2004.